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(54) **MULTI-BAND PATCH ANTENNA**

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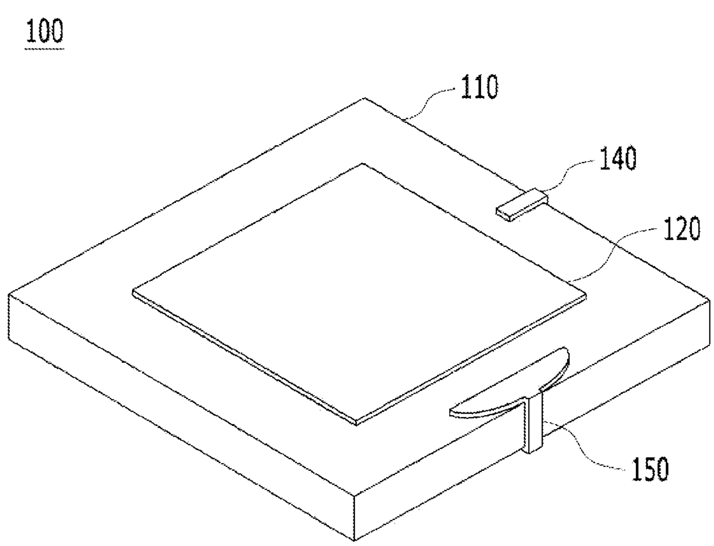
(57) **ABSTRACT**

Provided is a multi-band patch antenna which has, on a base substrate, a feeding patch and radiation patches, which are formed apart from an upper patch, so as to resonate with a first frequency band and a second frequency band. The provided multi-band patch antenna comprises: a base substrate having an upper surface, a lower surface, and a plurality of sides; an upper patch disposed on the upper surface of the base substrate; a lower patch disposed on the lower surface of the base substrate; a feeding patch disposed on the upper surface and a first side of the base substrate and disposed to be apart from the upper patch; and a first radiation patch disposed on the upper surface and a second side of the base substrate and disposed to be apart from the upper patch and the feeding patch on the upper surface of the base substrate.

10 Claims, 10 Drawing Sheets



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(58) **Field of Classification Search**
CPC H01Q 21/28; H01Q 9/0457; H01Q 9/0407; H01Q 9/045; H01Q 9/0421; H01Q 21/0025
See application file for complete search history.

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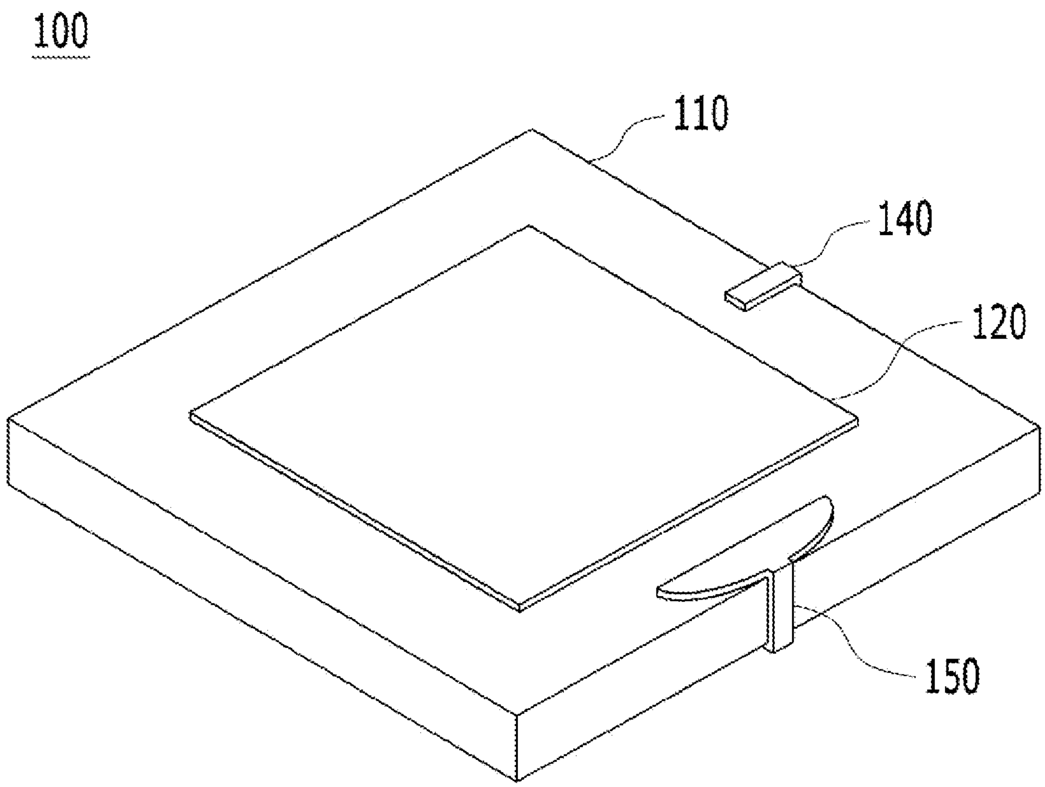
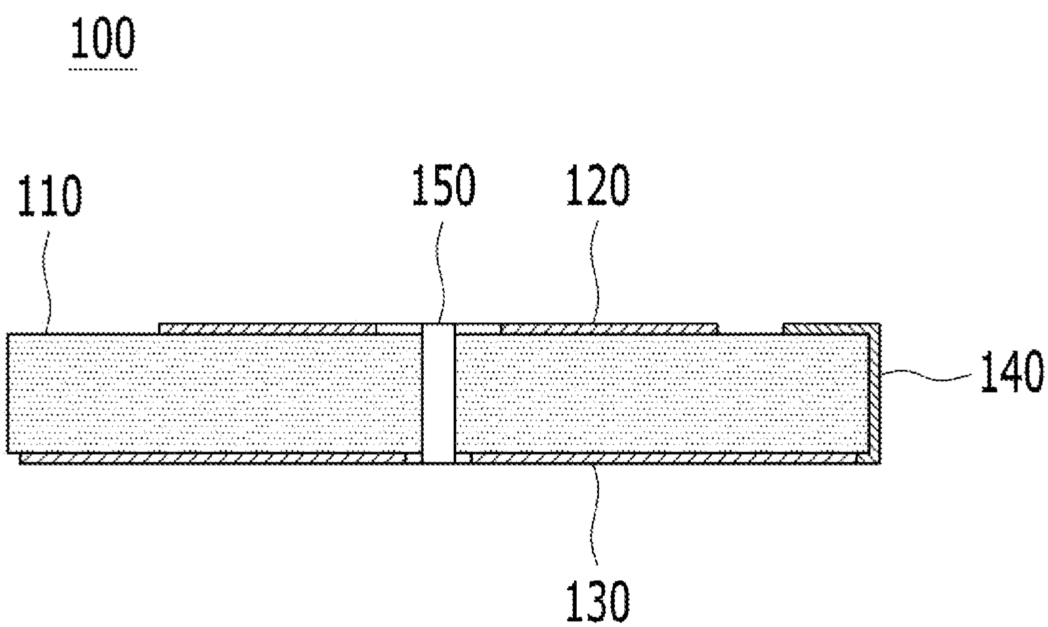
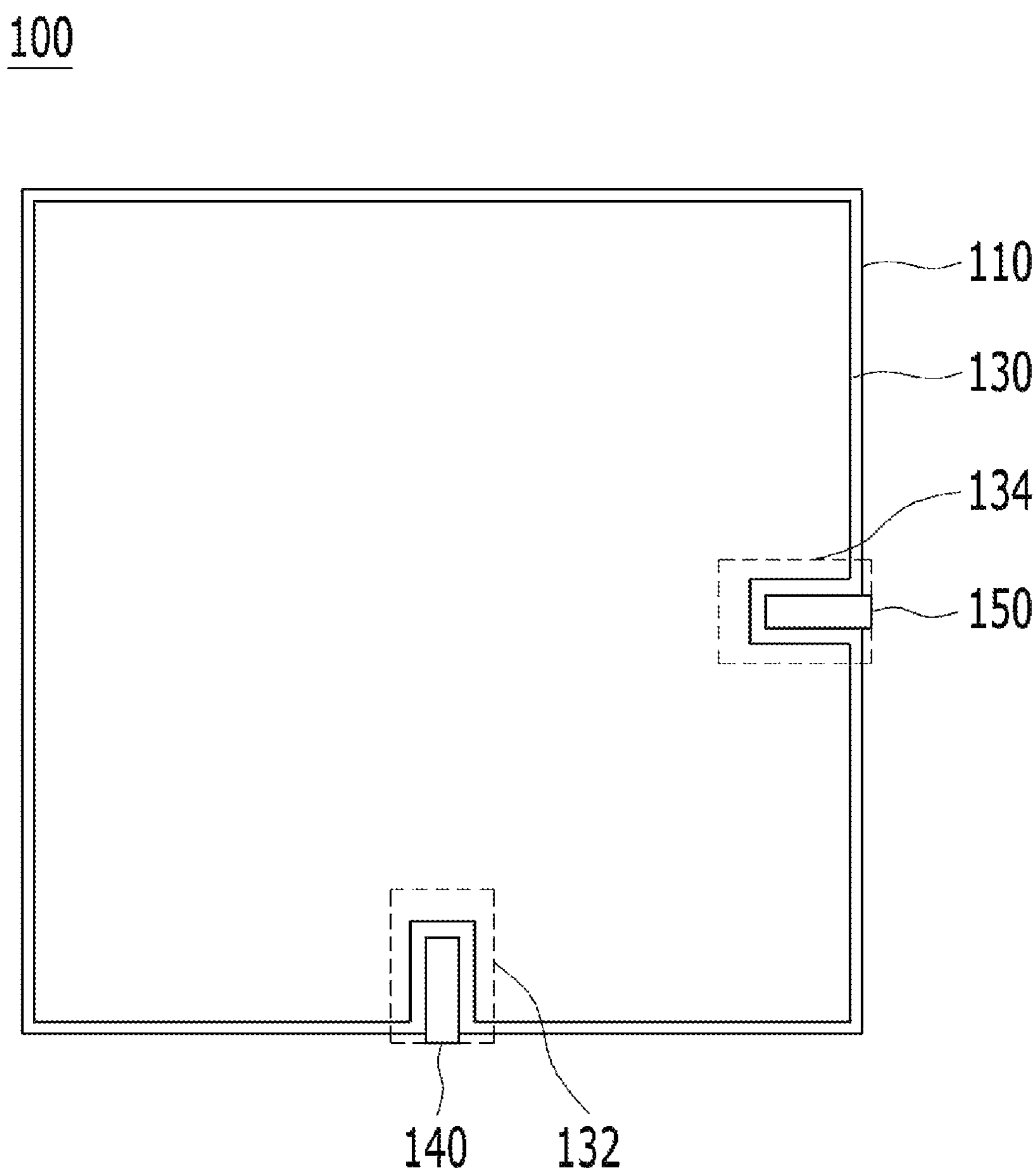
[FIG. 1]



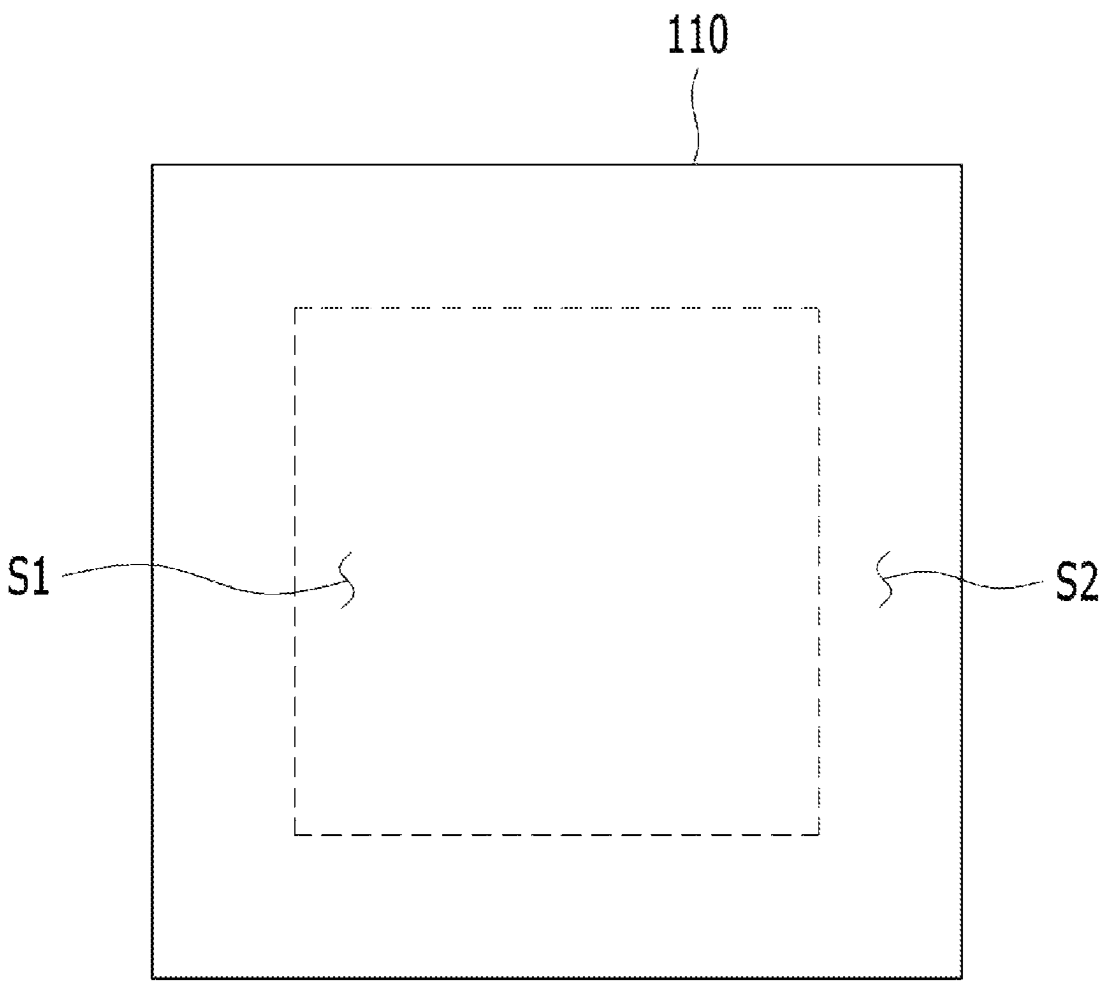
[FIG. 2]



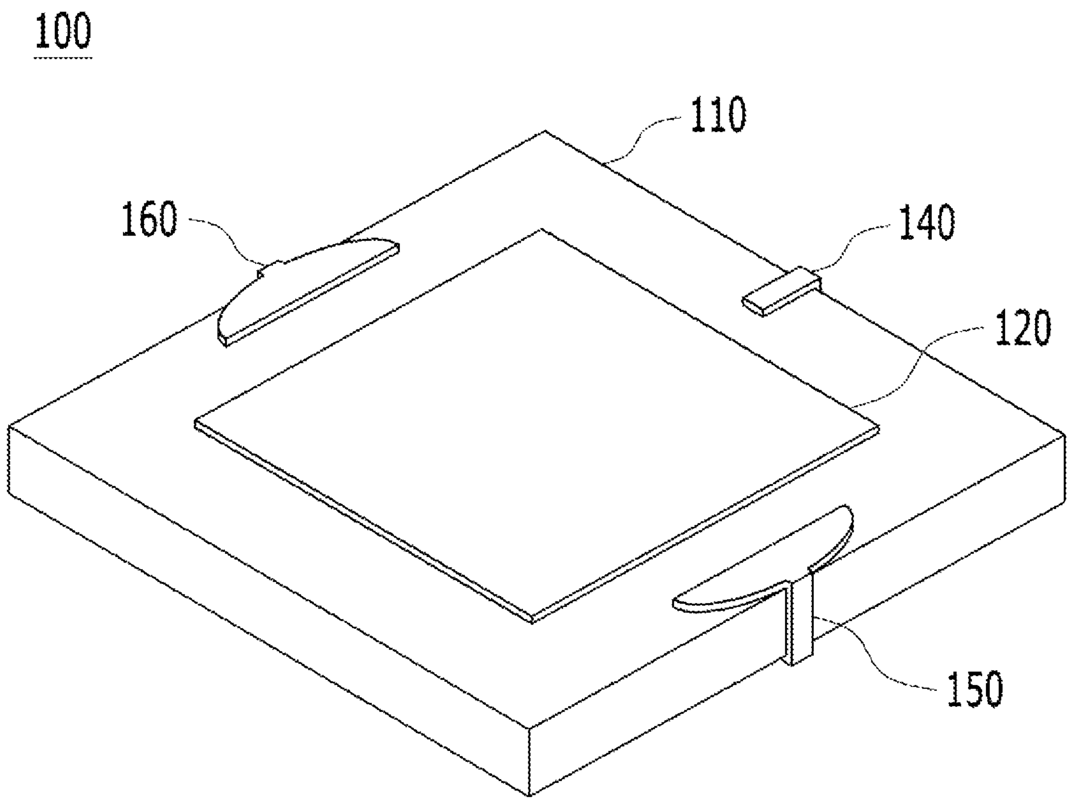
[FIG. 3]



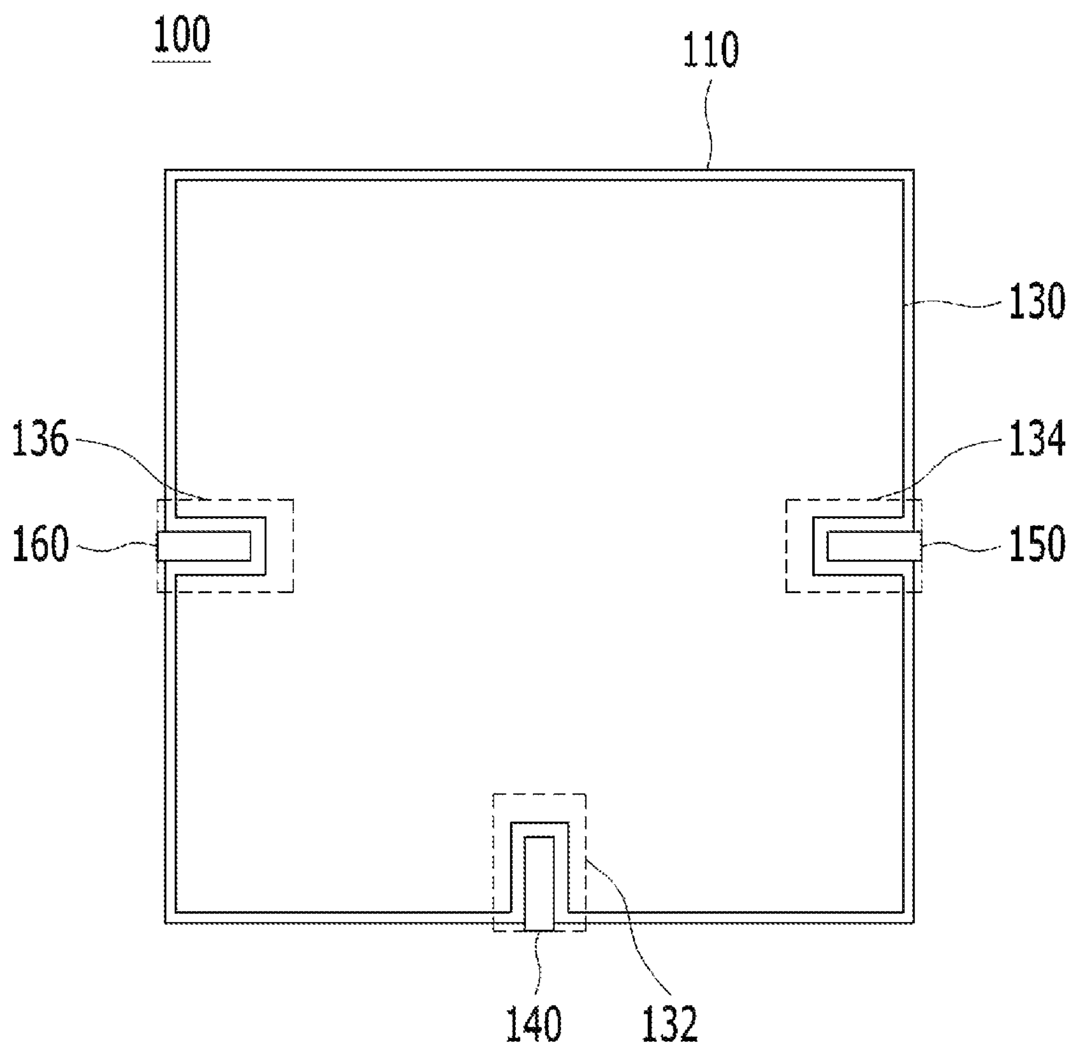
[FIG. 4]



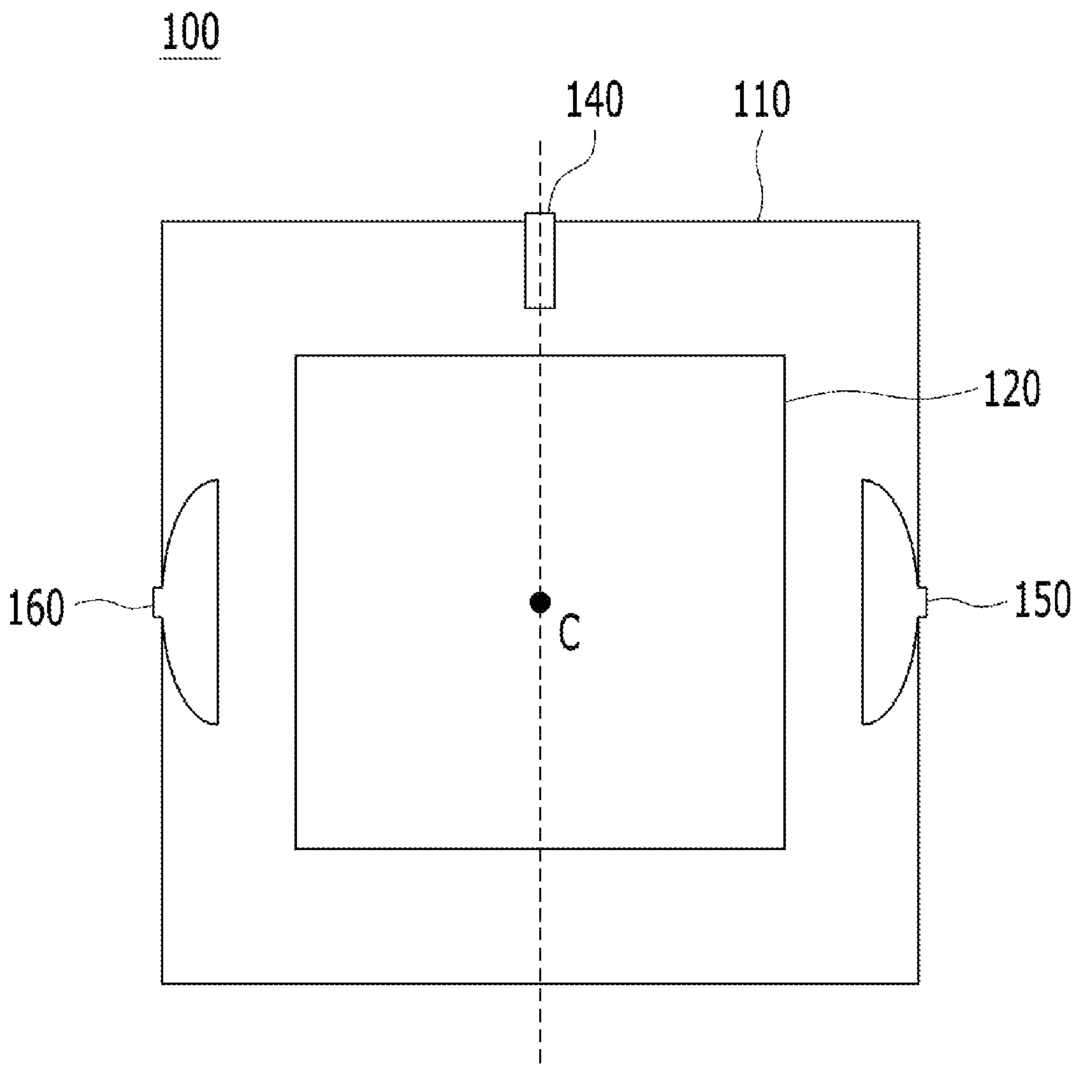
[FIG. 5]



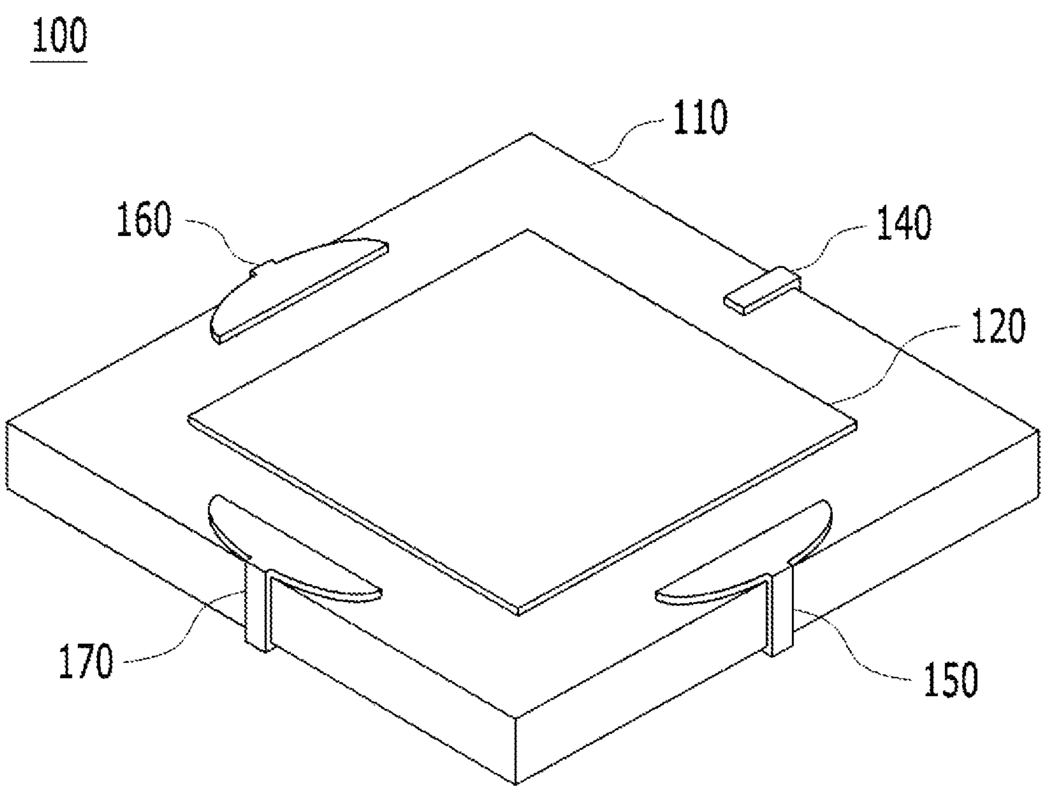
[FIG. 6]



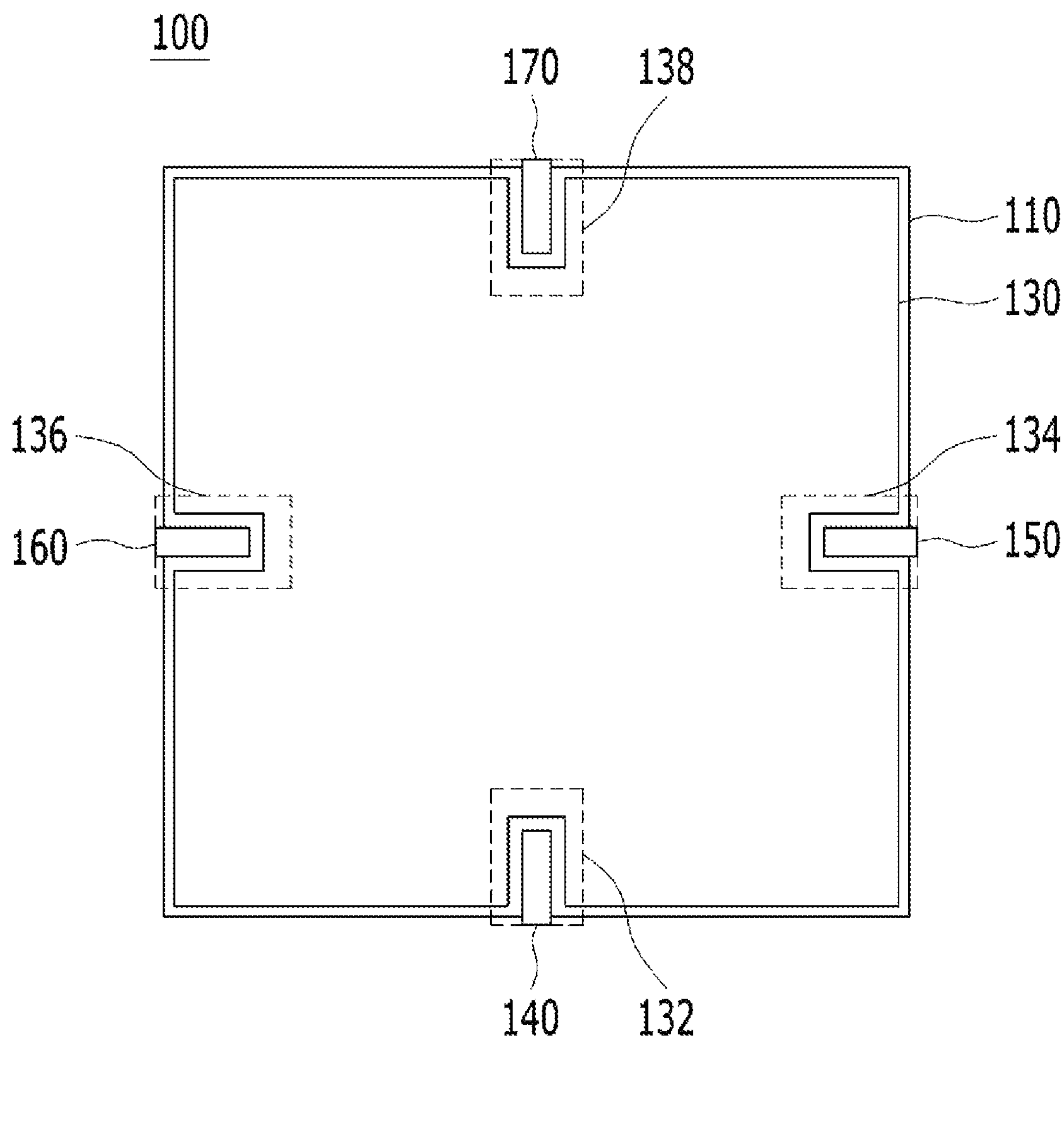
[FIG. 7]



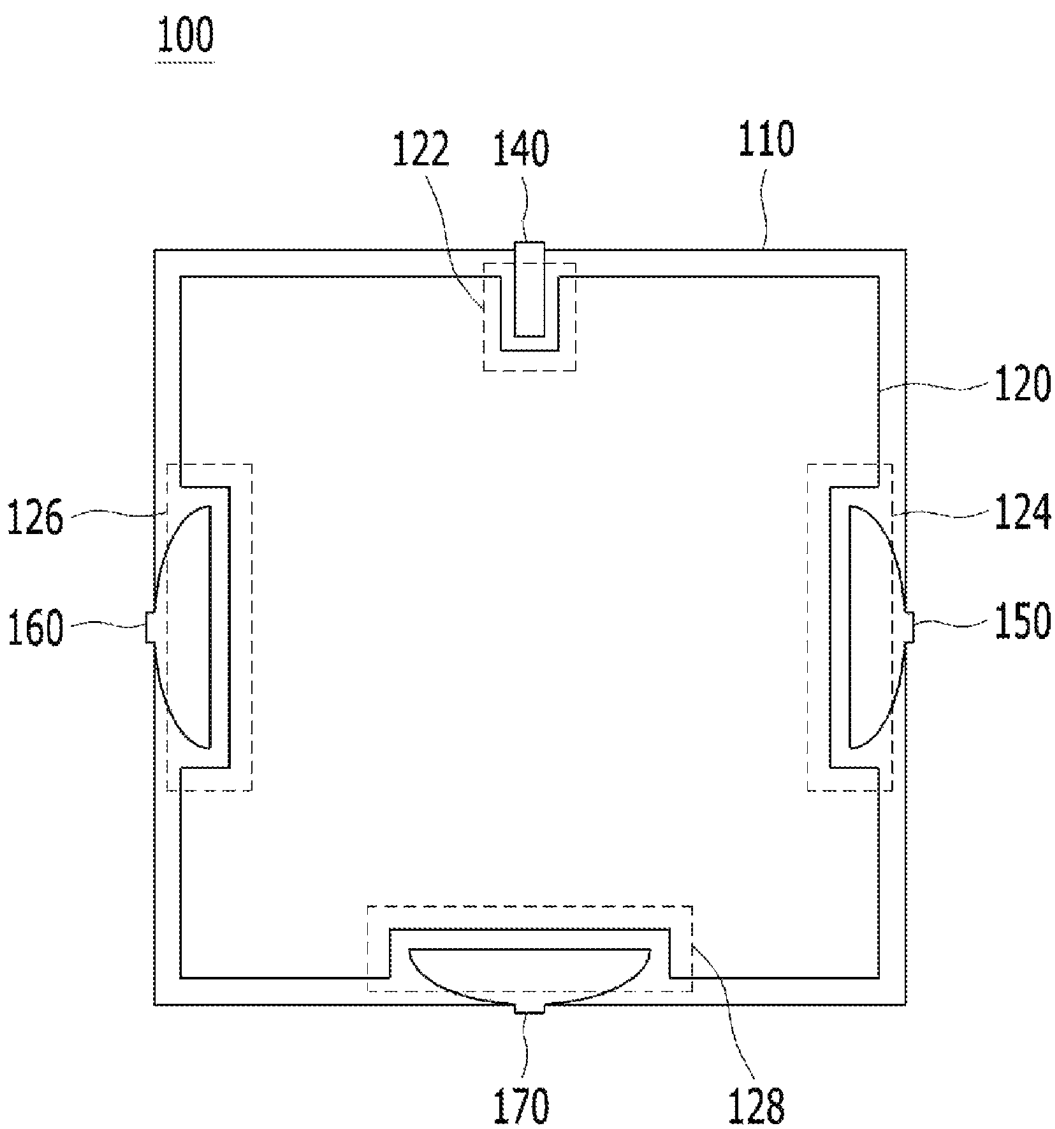
[FIG. 8]



[FIG. 9]



[FIG. 10]



MULTI-BAND PATCH ANTENNA

TECHNICAL FIELD

The present disclosure relates to a multi-band patch antenna and, more particularly, to a multi-band patch antenna that resonates to a frequency in a bandwidth of at least one of GPS, GLONASS, and SDARS and to a frequency in a bandwidth of Ultra-Wideband (UWB).

BACKGROUND ART

Usually, a patch antenna is used as an antenna that resonates to a frequency in a bandwidth of GNSS (for example, GPS (USA) or GLONASS (Russia)), SDARS (SiriusXM) or the like.

In recent years, with the advances in autonomous vehicular traveling, there has been a demand for higher-precision positional information. Accordingly, there has been a demand for an antenna that resonates to a frequency in a bandwidth at which high-precision positional information can be transmitted as in UWB, BLE, WiFi, and the like.

SUMMARY OF INVENTION

Technical Problem

An object of the present disclosure, which is made in view of the above-mentioned situation, is to provide a multi-band patch antenna in which a feeding patch and one or more radiation patches are formed on a base substrate in such a manner as to be spaced apart from an upper patch and which resonates to a frequency in a second bandwidth, as well as to a frequency in a first bandwidth that is a frequency bandwidth of GNSS.

Solution to Problem

In order to accomplish the above-mentioned object, according to an aspect of the present disclosure, there is provided a multi-band patch antenna including: a base substrate having an upper surface, a lower surface, and a plurality of lateral surfaces; an upper patch arranged on the upper surface of the base substrate; a lower patch arranged on the lower surface of the base substrate; a feeding patch arranged in a manner that comes into contact with the upper surface of the base substrate and a first lateral surface thereof and is spaced apart from the upper patch on the upper surface of the base substrate; and a first radiation patch arranged in a manner that comes into contact with the upper surface of the base substrate and a second lateral surface thereof and is spaced apart from the upper patch and the feeding patch on the upper surface of the base substrate.

In the multi-band patch antenna, the upper surface of the base substrate may be divided into a first region on which the upper patch is arranged and a second region on which the upper patch is not arranged, and a first end portion of the feeding patch and a first end portion of the first radiation patch may be formed on the second region of the base substrate in a manner that is spaced apart from the upper patch.

In the multi-band patch antenna, a second end portion of the feeding patch and a second end portion of the first radiation patch may be arranged on the lower surface of the base substrate, a first accommodation in which a second end portion of the feeding patch is accommodated and a second accommodation groove in which a second end portion of the first radiation patch is accommodated may be defined on the lower patch, and the feeding patch and the first radiation patch may be arranged in such a manner as to be spaced apart from the lower patch.

The multi-band patch antenna may further include a second radiation patch arranged in a manner that comes into contact with the upper surface of the base substrate and a third lateral surface thereof and is spaced apart from the upper patch and the feeding patch on the upper surface of the base substrate. In the multi-band patch antenna, the second radiation patch may be formed in such a manner that a first end portion thereof is arranged on the second region of the base substrate and is spaced apart from the upper patch and that a second end portion thereof is arranged on the lower surface of the base substrate, and a third accommodation groove in which the second end portion of the second radiation patch is accommodated may be defined on the lower patch.

The multi-band patch antenna may further include a third radiation patch arranged in a manner that comes into contact with the upper surface of the base substrate and a fourth lateral surface thereof and is spaced apart from the upper patch and the feeding patch on the upper surface of the base substrate. In the multi-band patch antenna, the third radiation patch may be formed in such a manner that a first end portion thereof is arranged on the second region of the base substrate and is spaced apart from the upper patch and that a second end portion thereof is arranged on the lower surface of the base substrate and a fourth accommodation groove in which the second end portion of the third radiation patch is accommodated may be defined on the lower patch.

In the multi-band patch antenna, a fifth accommodation groove in which the first end portion of the feeding patch is accommodated and a sixth accommodation groove in which the first end portion of the first radiation patch is accommodated may be defined on the upper patch.

Advantageous Effects of Invention

In a multi-band patch antenna according to the present disclosure, an additional radiation pattern is integrated into the existing patch antenna structure. As a result, the multi-band patch antenna can function as an outdoor positioning antenna that receives signals from a satellite outdoors. A composite antenna, capable of indoors and outdoors performing both indoor and outdoor positioning, can be implemented with a simple structure, using a UWB antenna.

In addition, unlike in a patch antenna in the related art where an additional patch needs to be formed or patches need to be stacked on top of each other, the simple addition of the radiation pattern in the multi-band patch antenna can create a patch antenna that resonates to frequencies in two different bandwidths, while minimizing any increase in size.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a perspective view illustrating a multi-band patch antenna according to a first embodiment of the present disclosure.

FIG. **2** is a front view illustrating the multi-band patch antenna according to the first embodiment of the present disclosure.

FIG. **3** is a bottom view illustrating the multi-band patch antenna according to the first embodiment of the present disclosure.

FIG. **4** is a view that is referred to for description of regional division of a base substrate in FIG. **1**.

FIG. **5** is a perspective view illustrating a multi-band patch antenna according to a second embodiment of the present disclosure.

FIG. **6** is a bottom view illustrating the multi-band patch antenna according to the second embodiment of the present disclosure.

FIG. **7** is a top view illustrating the multi-band patch antenna according to the second embodiment of the present disclosure.

FIG. **8** is a perspective view illustrating a multi-band patch antenna according to a third embodiment of the present disclosure.

FIG. **9** is a bottom view illustrating the multi-band patch antenna according to the third embodiment of the present disclosure.

FIG. **10** is a view that is referred to for description of a modification example of the multi-band patch antenna to the embodiment of the present disclosure.

DESCRIPTION OF EMBODIMENTS

Preferred embodiments of the present disclosure will be in detail described with reference to the accompanying drawings in such a manner as to enable an ordinary person of ordinary skill in the art to which the present disclosure pertains to practice the present disclosure without undue experimentation. It should be noted that the same constituent elements, although illustrated in different drawings, are designated by the same reference numeral when a reference numeral is assigned to a constituent element in the drawings. In addition, a specific description of a well-known configuration or function that is associated with the present disclosure will be omitted when determined as making the nature and gist of the present disclosure obfuscated.

With reference to FIGS. **1** to **3**, a multi-band patch antenna **100** according to a first embodiment of the present disclosure is configured to include a base substrate **110**, an upper patch **120**, a lower patch **130**, a feeding pattern **140**, and a first radiation pattern **150**.

The base substrate **110** is configured as a dielectric component having an upper surface, a lower surface, and a plurality of lateral surfaces. As an example, the base substrate **110** is configured as a dielectric substrate formed of a ceramic material that is characterized by a high permittivity, a low thermal expansion coefficient, and the like.

The base substrate **110** may be configured as a magnetic body having an upper surface, a lower surface, and a plurality of lateral surfaces. As an example, the base substrate **110** is configured as a magnetic substrate formed from a ferrite magnet or the like.

With reference to FIG. **4**, as an example, the base substrate **110** includes a first lateral surface, a second lateral surface facing the first lateral surface, a third lateral surface connected to a first end portion of the first lateral surface and a first end portion of the second lateral surface, and a fourth surface connected to a second end portion of the first lateral surface and a second end portion of the second lateral surface in a manner that faces the third lateral surface. Furthermore, an upper surface of the base substrate **110** is divided into a first region **S1** in which the upper patch **120** is arranged, and a second region **S2** in which the upper patch **120** is not arranged.

The upper patch **120** is arranged on an upper surface of the base substrate **110**. The upper patch **120** is arranged on the upper surface of the base substrate **110**, but on the first region **S1** of the base substrate **110**.

The upper patch **120** is made from a thin plate formed of a conductive material, such as copper, aluminum, gold, or silver, that has a high electric conductivity. The upper patch **120** may be formed to have a horizontal cross section with various shapes, such as a rectangle, a triangle, and an octagon, according to a shape of the base substrate **110**.

The upper patch **120** may be changed to various shapes through a process, such as frequency tuning. In this point, the upper patch **120** is fed through the feeding pattern **140** and thus operates as an antenna that resonates to a frequency in a bandwidth of one of GNSS (for example, GPS (USA) or GLONASS (Russia)) and SDARS (for example, SiriusXM).

The lower patch **130** is arranged on a lower surface of the base substrate **110**. The lower patch **130** is made from a thin plate formed of a conductive material, such as copper, aluminum, gold, or silver, that has a high electric conductivity. The lower patch **130** may be formed to have a horizontal cross section with various shapes, such as a rectangle, a triangle, and an octagon, according to the shape of the base substrate **110**. In this case, as an example, the lower patch **130** is a patch for grounding (GND).

A first accommodation groove **132** in which a second end portion of the feeding pattern **140** is accommodated and second accommodation groove **134** in which a second end portion of the first radiation pattern **150** is accommodated may be formed in the lower patch **130**.

The first accommodation groove **132** and the second accommodation groove **134** are firmed by cutting off one portion of the lower patch **130** in a manner that extends from an edge of the lower patch **130** toward the direction of the center point of the lower patch **130**.

The first accommodation groove **132** may be formed to have a horizontal cross section with various shapes, such as a circle, a rectangle, a triangle, and a pentagon. The first accommodation groove **132** may have any shape that allows for accommodation of one portion (one portion, to the side of the second end portion, of the feeding pattern **140**) of the feeding pattern **140**.

The second accommodation groove **134** may be formed to have a horizontal cross section with various shapes, such as a circle, a rectangle, a triangle, and a pentagon. The second accommodation groove **134** may have any shape that allows for accommodation of one portion (one portion, to the side of the second end portion, of the first radiation pattern **150**) of the first radiation pattern **150**.

The feeding pattern **140** feeds the upper patch **120** in order to operate the upper patch **120** as a first antenna. The feeding pattern **140** is arranged in a manner that comes into contact with the upper surface, the lateral surface, and the lower surface of the base substrate **110**. In this case, the feeding pattern **140** is arranged on one of the first to fourth lateral surfaces of the base substrate **110**.

The feeding pattern **140** is formed in such a manner that a first end portion thereof is arranged on the upper surface of the base substrate **110**, but on the second region **S2** of the base substrate **110** in a manner that is spaced a predetermined distance apart from the upper patch **120**. The first end portion of the feeding pattern **140** is arranged on the upper surface of the base substrate **110** in a manner that is spaced a predetermined distance apart from the upper patch **120** and thus forms a coupling feeding structure along with the upper patch **120**.

The feeding pattern **140** is arranged in a manner that comes into contact with the upper surface and the lateral surface of the base substrate **110**. The feeding pattern **140** extends from the upper surface of the base substrate **110** to the lateral surface thereof. Thus, the second end portion of the feeding pattern **140** is arranged on one lateral surface of the base substrate **110**.

The feeding pattern **140** may extend from the upper surface of the base substrate **110** through the lateral surface thereof up to the lower surface thereof in a manner that comes into contact with the upper, lateral, and lower surfaces of the base substrate **110**. Thus, the second end portion of the feeding pattern **140** may be arranged on the lower surface of the base substrate **110**. In this case, the feeding pattern **140** is arranged in such a manner that the second end portion thereof is accommodated in the first accommodation groove **132** in the lower patch **130** and is spaced a predetermined distance apart from the lower patch **130**.

The first radiation pattern **150** operates as a second antenna that resonates to a frequency in a different bandwidth than the upper patch **120**. In this case, as an example, the second antenna is an antenna that resonates to a signal in a UWB frequency bandwidth.

The first radiation pattern **150**, made from a thin metal plate, is arranged on the second region S**2** of the base substrate **110**. As an example, the first radiation pattern **150** is made from a thin metal plate formed of a conductive material, such as copper, aluminum, gold, or silver, that has a high electric conductivity.

The first radiation pattern **150** is formed in such a manner that a first end portion thereof is arranged on the upper surface of the base substrate **110**, but on the second region S**2** of the base substrate **110** in a manner that is spaced a predetermined distance apart from the upper patch **120**. In this case, the first end portion of the first radiation pattern **150** may be formed in the shape of a plate that has a horizontal cross section with various shapes, such as a circle, a semicircle, an ellipse, and a semi-ellipse.

The first radiation pattern **150** is arranged in a manner that comes into contact with the upper surface and the lateral surface of the base substrate **110**. The first radiation pattern **150** extends from the upper surface of the base substrate **110** to the lateral surface thereof. Thus, the second end portion of the first radiation pattern **150** is arranged on one lateral surface of the base substrate **110**.

The first radiation pattern **150** may extend from the upper surface of the base substrate **110** through the lateral surface thereof up to the lower surface thereof in a manner that comes into contact with the upper, lateral, and lower surfaces of the base substrate **110**. Thus, the second end portion of first radiation pattern **150** may be arranged on the lower surface of the base substrate **110**. In this case, the first radiation pattern **150** is arranged in such a manner that the second end portion thereof is accommodated in the second accommodation groove **134** in the lower patch **130** and is spaced a predetermined distance apart from the lower patch **130**.

The first radiation pattern **150** is arranged on the base substrate **110**, but in a manner that comes into contact with the lateral surface of the base substrate **110** on which the feeding pattern **140** is not formed. As an example, in a case where the feeding pattern **140** is arranged in a manner that comes into contact with the first lateral surface of the base substrate **110**, the first radiation pattern **150** is arranged in a manner that comes into contact with one of the second to fourth lateral surfaces of the base substrate **110**.

Withe reference to FIGS. **5** and **6**, a multi-band patch antenna **100** according to a second embodiment of the present disclosure may be configured to include a second radiation pattern **160** in addition to the constituent elements of the multi-band patch antenna **100** according to the first embodiment of the present disclosure.

The second radiation pattern **160** operates as the second antennas, together with the first radiation pattern **150**. The second radiation pattern **160**, made from a thin metal plate, is formed on the second region S**2** of the base substrate **110**. As an example, the second radiation pattern **160** is made from a thin metal plate formed of a conductive material, such as copper, aluminum, gold, or silver, that has a high electric conductivity.

The second radiation pattern **160** is formed in such a manner that a first end portion thereof is arranged on the upper surface of the base substrate **110**, but on the second region S**2** of the base substrate **110** in a manner that is spaced a predetermined distance apart from the upper patch **120**. In this case, the first end portion of the second radiation pattern **160** may be formed in the shape of a plate that has a horizontal cross section with various shapes, such as a circle, a semicircle, an ellipse, and a semi-ellipse.

The second radiation pattern **160** is arranged in a manner that comes into contact with the upper surface and the lateral surface of the base substrate **110**. The second radiation pattern **160** extends from the upper surface of the base substrate **110** to the lateral surface thereof. Thus, a second end portion of the second radiation pattern **160** is arranged on one lateral surface of the base substrate **110**.

The second radiation pattern **160** extends from the upper surface of the base substrate **110** through the lateral surface thereof up to the lower surface thereof in a manner that comes into contact with the upper, lateral, and lower surfaces of the base substrate **110**. Thus, the second end portion of the second radiation pattern **160** may be arranged on the lower surface of the base substrate **110**. In this case, a third accommodation groove **136** is further formed in the lower patch **130**. Furthermore, the second end portion of the second radiation pattern **160** is arranged in a manner that is accommodated in the third accommodation groove **136** in the lower patch **130** and is spaced a predetermined distance apart from the lower patch **130**.

In this case, the second radiation pattern **160** is arranged on the base substrate **110**, but in a manner that comes into contact with one lateral surface of the base substrate **110** on which the feeding pattern **140** and the first radiation pattern **150** are not formed. As an example, the feeding pattern **140** is arranged in a manner that comes into contact with the first lateral surface of the base substrate **110**, and the first radiation pattern **150** is arranged in a manner that comes into contact with one of the second to fourth lateral surfaces of the base substrate **110**. In this case, the second radiation pattern **160** is arranged in a manner that comes into contact with one, on which the first radiation pattern **150** is not formed, of the second to fourth lateral surfaces.

With reference to FIG. **7**, the second radiation pattern **160** is arranged in a manner that faces the first radiation pattern **150** with the center point C of the base substrate **110** in between. As an example, when the first radiation pattern **150** is arranged in a manner that contacts into contact with the third lateral surface of the base substrate **110**, the second radiation pattern **160** is arranged in a manner that comes into contact with the fourth lateral surface of the base substrate **110** and thus faces the first radiation pattern **150**.

With reference to FIGS. **8** and **9**, a multi-band patch antenna **100** according to a third embodiment of the present disclosure may be configured to include a third radiation pattern **170** in addition to the constituent elements of the multi-band patch antenna **100** according to the first embodiment of the present disclosure.

The third radiation pattern **170** operates as the second antenna, together with the first radiation pattern **150**. The third radiation pattern **170**, made from a thin metal plate, is formed on the second region S2 of the base substrate **110**. As an example, the third radiation pattern **170** is made from a thin metal plate formed of a conductive material, such as copper, aluminum, gold, or silver, that has a high electric conductivity.

The third radiation pattern **170** is formed in such a manner that a first end portion thereof is arranged on the upper surface of the base substrate **110**, but on the second region S2 of the base substrate **110** in a manner that is spaced a predetermined distance apart from the upper patch **120**. In this case, the first end portion of the third radiation pattern **170** may be formed in the shape of a plate that has a horizontal cross section with various shapes, such as a circle, a semicircle, an ellipse, and a semi-ellipse.

The third radiation pattern **170** is arranged in a manner that comes into contact with the upper surface and the lateral surface of the base substrate **110**. The second radiation pattern **160** extends from the upper surface of the base substrate **110** toward the lateral surface thereof. Thus, a second end portion of the third radiation pattern **170** is arranged on one lateral surface of the base substrate **110**.

The third radiation pattern **170** extends from the upper surface of the base substrate **110** through the lateral surface thereof up to the lower surface thereof in a manner that comes into contact with the upper, lateral, and lower surfaces of the base substrate **110**. Thus, the second end portion of the third radiation pattern **170** may be arranged on the lower surface of the base substrate **110**. In this case, a fourth accommodation groove **138** is further formed in the lower patch **130**. The second end portion of the third radiation pattern **170** is arranged in a manner that is accommodated in a fourth accommodation groove **138** in the lower patch **130** and is spaced a predetermined distance apart from the lower patch **130**.

The third radiation pattern **170** is formed on one lateral surface, on which the feeding pattern **140**, the first radiation pattern **150**, and the second radiation pattern **160** are not formed, of the base substrate **110**. As an example, the feeding pattern **140** is arranged in a manner that comes into contact with the first lateral surface of the base substrate **110**, and the first radiation pattern **150** and the second radiation pattern **160** are arranged in such a manner as to come into contact with the third and fourth lateral surfaces, respectively, of the base substrate **110**. In this case, the third radiation pattern **170** is arranged to extend from the second lateral surface of the base substrate **110** up to the fourth lateral surface thereof in a manner that comes into contact with the second lateral surface of the base substrate **110** to the fourth lateral surfaces thereof on which the 1+first radiation pattern **150** is not formed.

Reduction in size of the multi-band patch antenna **100** results in reduction in size of the second region S2 of the base substrate **110**. For this reason, the feeding pattern **140** and the radiation pattern (that is, at least one of the first radiation pattern **150**, the second radiation pattern **160**, and the third radiation pattern **170**) are electrically connected to the upper patch **120**, or interference occurs between the feeding pattern **140** and the radiation pattern. Thus, antenna performance decreases, or the second antenna cannot be configured.

In order to address this problem, there is a need to maintain a separation distance between each of the feeding pattern **140** and the radiation pattern and the upper patch **120**. To this end, with reference to FIG. **10**, a fifth accommodation groove **122** in which the feeding pattern **140** is accommodated, a sixth accommodation groove **124** in which the first radiation pattern **150** is accommodated, a seventh accommodation groove **126** in which the second radiation pattern **160** is accommodated, and an eighth accommodation groove **128** in which the third radiation pattern **170** is accommodated may be formed in the upper patch **120**.

Each of the fifth to eighth accommodation grooves **122** to **128** is formed by cutting off one portion of the upper patch **120**, but in the direction from an edge of the upper patch **120** to the center of the upper patch **120**. As an example, the fifth to eighth accommodation grooves **122** to **128** may be formed to have a horizontal cross section with various shapes, such as a circle, a rectangle, a triangle, and a pentagon. The fifth to eighth accommodation grooves **122** to **128** may have any shape that allows for accommodation of one portion of the feeding pattern **140** and the radiation pattern.

FIG. **10** illustrates that four accommodation grooves are formed in the upper patch **120**. However, the upper patch **120** is not limited to the four accommodation grooves. One, two, or three accommodation grooves may be formed in the upper patch **120**, depending on the number of the radiation patterns. That is, in a case where one radiation pattern is provided, two accommodation grooves (that is, the fifth accommodation groove **122** and the sixth accommodation groove **124**) are formed in the upper patch **120**. That is, in a case where two radiation patterns are provided, three accommodation grooves (that is, the fifth accommodation groove **122**, the sixth accommodation groove **124**, and the seventh accommodation groove **126**) are formed in the upper patch **120**. That is, in a case where three radiation patterns are provided, four accommodation grooves (that is, the fifth accommodation groove **122**, the sixth accommodation groove **124**, the seventh accommodation groove **126**, and the eighth accommodation groove **128**) are formed in the upper patch **120**.

The preferred embodiments of the present disclosure are described above. However, the present disclosure may be practiced in various forms. It would be apparent to a person of ordinary skill in the art that various modifications and alterations may be made to the preferred embodiments without departing from the scope of the claims of the present disclosure.

The invention claimed is:

1. A multi-band patch antenna comprising:

a base substrate having an upper surface, a lower surface, and a plurality of lateral surfaces;

an upper patch arranged on the upper surface of the base substrate;

a lower patch arranged on the lower surface of the base substrate;

a feeding patch arranged on the upper surface of the base substrate and a first lateral surface thereof, and spaced apart from the upper patch on the upper surface of the base substrate; and

a first radiation patch arranged on the upper surface of the base substrate and a second lateral surface thereof, and spaced apart from the upper patch and the feeding patch on the upper surface of the base substrate, and

wherein an accommodation groove is formed in the upper patch to accommodate a portion of the feeding patch arranged on the upper surface of the base substrate, and another accommodation groove is formed in the upper patch to accommodate a portion of the first radiation patch arranged on the upper surface of the base substrate.

2. The multi-band patch antenna of claim 1, wherein the upper surface of the base substrate is divided into a first region on which the upper patch is arranged and a second region on which the upper patch is not arranged.

3. The multi-band patch antenna of claim 2, wherein the feeding patch and the first radiation patch are formed so that a first end portion of the feeding patch and a first end portion of the first radiation patch are arranged on the second region of the base substrate to be spaced apart from the upper patch.

4. The multi-band patch antenna of claim 2, wherein a second end portion of the feeding patch and a second end portion of the first radiation patch are arranged on the lower surface of the base substrate.

5. The multi-band patch antenna of claim 4, wherein a first accommodation groove in which the second end portion of the feeding patch is accommodated and a second accommodation groove in which the second end portion of the first radiation patch is accommodated are defined on the lower patch, and

wherein the feeding patch and the first radiation patch are arranged in such a manner as to be spaced apart from the lower patch.

6. The multi-band patch antenna of claim 4, further comprising:

a second radiation patch arranged on the upper surface of the base substrate and a third lateral surface thereof, and spaced apart from the upper patch and the feeding patch on the upper surface of the base substrate.

7. The multi-band patch antenna of claim 6, wherein the second radiation patch is formed so that a first end portion thereof is spaced apart from the upper patch on a second region of the base substrate and that a second end portion of thereof is arranged on the lower surface of the base substrate, and

wherein a third accommodation groove in which the second end portion of the second radiation patch is accommodated is defined on the lower patch.

8. The multi-band patch antenna of claim 6, further comprising:

a third radiation patch arranged on the upper surface of the base substrate and a fourth lateral surface thereof and spaced apart from the upper patch and the feeding patch on the upper surface of the base substrate.

9. The multi-band patch antenna of claim 8, wherein the third radiation patch is formed so that a first end portion thereof is arranged on a second region of the base substrate and is spaced apart from the upper patch and that a second end portion thereof is arranged on the lower surface of the base substrate, and

wherein a fourth accommodation groove in which the second end portion of the third radiation patch is accommodated is defined on the lower patch.

10. The multi-band patch antenna of claim 1, wherein a fifth accommodation groove in which a first end portion of the feeding patch is accommodated and a sixth accommodation groove in which a first end portion of the first radiation patch is accommodated are defined on the upper patch.

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